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Occurrence of fatigue over 20 years after recovery from Guillain–Barré syndrome

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ABSTRACT

Introduction: Reports regarding outcome of Guillain–Barré syndrome (GBS) are usually limited to relatively short follow-up. We assessed the occurrence of fatigue and correlated it to clinical measures in patients who had suffered from GBS over 20 years previously.
Methods: We contacted 24 patients with GBS requiring in-hospital rehabilitation during the years 1970–1987. Disability was established at rehabilitation admission using the Hughes scale (HS). Disability and fatigue were assessed at time of this study by the Overall Disability Sum Score (ODSS) and the Fatigue Severity Scale (FSS).
Results: Mean HS at admission was 2.5 ± 0.7, and at follow-up 1.1 ± 1.3. Mean ODSS was 2.8 ± 3.6, FSS was 4.4 ± 2. Ten patients reported severe fatigue (>5). Very good correlations were found between FSS and HS at admission and follow-up and between FSS and ODSS. FSS was not influenced by patients' age, age at disease onset, gender or time from GBS to the study.
Conclusion: Fatigue can persist after apparent recovery from GBS and remain severe for many years.

1. Introduction

Guillain–Barré syndrome (GBS) is an acute demyelinating polyradiculoneuropathy that leads to limb and trunk weakness, and, if severe, also weakness of respiratory and bulbar muscles. It can affect individuals at any age, although it is more frequent in the elderly, with an annual incidence of approximately 1.2–2.3 cases per 100 000 persons [1,2]. The medical condition improves in most patients, with an overall survival exceeding 90%. Residual motor disability of different degrees is reported in 10–20% of patients [3].

Fatigue is a frequent complaint in the general practitioners' practice. The pathophysiologic basis of fatigue is complex and includes physical and psychological aspects. It is an important feature of many neurological diseases, such as multiple sclerosis and Parkinson's disease, as well as a presentation of depression and a common side effect of medications, e.g. beta-blockers [4].

In GBS fatigue was described in 60–80% of patients [5,6]. Even in patients who achieved a good functional recovery, fatigue is reported as one of the most disabling residual symptoms [6]. It is not clear, however, for how long symptoms of fatigue persist after recovery from acute GBS and whether there are any factors that might influence the occurrence and the magnitude of the residual fatigue after GBS. Previous studies that analyzed the relationship between GBS and fatigue evaluated residual fatigue relatively early after recovery from the acute disease and the long-term prognosis remains largely unknown.

The purpose of the present study was to identify patients who suffered an acute episode of GBS 20–35 years previously in order to evaluate whether patients who apparently recovered from the acute illness report increased fatigue and whether the amount of fatigue reported is related to the severity of the disease in the acute phase, to the residual motor deficit or other individual variables, such as age and gender.

2. Patients and methods

We retrieved the records of all patients with a diagnosis of GBS who were admitted for in-hospital rehabilitation during the years 1970–1987 at a large rehabilitation clinic. Patients referred for in-hospital rehabilitation are a priori a selected group with more disabling weakness during the acute phase. Noticeably this population did not receive any immunomodulatory treatment, as those treatments were not yet used at the time of their hospitalization, and thus reflects the natural history of the disease. The clinical diagnosis was re-evaluated retrospectively using accepted modern clinical and electrophysiological criteria [7]. Patients with a confirmed diagnosis of GBS were contacted by phone.

Patients, in whom the diagnosis of GBS was not confirmed at re-evaluation of the charts, died or were lost for follow-up, those who reported during the phone interview to suffer from significant

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systemic, oncologic or additional neurologic diseases or were taking drugs that might cause fatigue, were not included in the study.

In all patients age at study evaluation and at acute GBS, gender, and duration since the hospitalization to the present study were recorded. The severity of disease at admission to the rehabilitation clinic was estimated retrospectively using the Hughes scale (HS) [8], since the information needed for this evaluation was available for all patients and the scoring criteria are straightforward. The HS score ranges from 1 to 5.

Patients were then interviewed by phone by a physician experienced in neurological rehabilitation. HS at time of the interview was determined. The disability at the time of the present study was evaluated by the Overall Disability Sum Score (ODSS) [9]. The ODSS is a questionnaire that evaluates functional disability of upper and lower limbs during daily activities, with a scale of 0–5 for the upper limbs and 0–7 for the lower limbs. Therefore the total ODSS score has a range from 0 (no disability) to 12 (severe disability, preventing all purposeful movements of all limbs). ODSS was used to evaluate disability at time of the study because it has a wider range of variability and therefore allows more accurate evaluation.

The fatigue of the patients was evaluated by the Fatigue Severity Score scale (FSS). FSS is a questionnaire that comprises 9 sentences regarding the subjective evaluation of fatigue. Each sentence is graded 1–7, and all responses are averaged to a score between 1(no fatigue) and 7 (most disabling fatigue) [10]. An FSS score of >5 is indicative of severe fatigue [6].

The study protocol was approved by the Institutional Review Board and an informed consent was signed by all study participants and mailed to the investigators.

Statistical evaluation of data was performed using Pearson correlation tests for comparisons between FSS and ODSS, age at disease onset, age at current evaluation and duration of follow-up respectively. Spearman correlations were performed in order to analyze the relationship between FSS and HS, as well as between ODSS and age at acute disease, age at current evaluation, HS, and duration of follow-up respectively. Mann–Whitney tests were performed for analysis of the relationship between patients' gender and FSS and ODSS respectively. $p<0.05$ was considered significant for all comparisons.

3. Results

We were able to identify hospitalization charts of 49 patients with a discharge diagnosis of GBS during the period 1970–1987. Among them 9 have since died, 14 patients were lost for follow-up, 2 were excluded due to language difficulties that rendered it impossible to answer the questionnaires. Therefore 24 patients (15 males, 62%) are included in the current study. Their mean age at participation in the present study was 57 ± 12 years (range 31–78 years, median 58). The age at acute hospitalization was 30 ± 12 years (range 11–50, median 28.5), while 5 patients were younger than 20 years at the time of acute GBS (mean 13, 4 males). The acute hospitalization for GBS occurred 27 ± 5 years before the present study (range 20–38, median 26.5). The HS of the patients during the acute hospitalization was estimated at 2.5 ± 0.7 (range 2–4, from patients able to walk without support to bedbound, median 2).

After discharge from the rehabilitation center, none of the interviewed patients reported a relapse of GBS, a late worsening of weakness or increased disability. The mean remaining HS at time of the study in our patient group was 1.1 ± 1.3 (range 0–4, median 1), in concordance with the expected natural improvement ($p<0.0001$ for the HS difference between the examinations). The mean ODSS was 2.8 ± 3.6 (range 0–10, median 1), mean FSS was 4.4 ± 2.0 (range 1.2–7, median 4.3). In 10 patients (42%) the FSS indicated a disabling level of fatigue (>5).

No significant correlation was found between patients' disability at the time of the study, as measured by the ODSS, or the level of daily fatigue, as measured by the FSS, and the age of patients at acute disease or follow-up, the time elapsed between the acute disease and the present study and patients' gender, respectively. However, a strong positive correlation was found between HS during the acute hospitalization and the present disability level (ODSS) ($p<0.0001$, $r=0.73$, Spearman correlation coefficient). Similarly, there was a strong positive correlation between HS at hospitalization and at follow-up and FSS ($p<0.001$, $r=0.74$, Spearman correlation coefficient) and between ODSS and FSS as well ($p<0.0001$, $r=0.79$, Pearson correlation coefficient) (Table 1).

The subgroup of five patients who suffered the acute GBS attack as children and adolescents (age<20) did not show any difference as compared to adult patients in any of the measured variables.

4. Discussion

GBS is considered one of the few neurological diseases with a very favorable prognosis, in spite of severe injury during the acute stage. Most patients improve markedly within weeks to months after the acute attack and are able to regain an active life. Until recently, the medical literature did not pay much attention to patients' complaints regarding impaired social functioning and quality of life after recovery from the acute phase of GBS. This is probably related to the relatively well-recovered muscle strength at the formal neurological examination in most of these patients, which creates expectations for good overall performance in them.

Lately several studies stressed the central role of fatigue in the subjective limitation of daily activities, in spite of good recovery of muscle strength. For example, Bernsen et al [11] examined the functional handicap during daily life activities in 122 patients following GBS, 3–6 years after the acute disease, and reported that approximately 40% of the patients had to change their work or life style due to residual functional disability. In a similar study [12] among 70 patients examined 3–5 years after GBS, 27% reported changes in personal life style because of increased fatigability.

Dornonville et al [3] and Khan et al [13] evaluated patients a mean of 6 years following the acute disease (range 1–14 years). In spite of good recovery of objective measures of motor skills, the quality of life of the patients showed prominent impairment as compared with healthy individuals in items related to capabilities at workplace and maintenance of social relationships, as well as in self-confidence and ability to live independently. Therefore, it seems that even after apparent recovery, GBS has a greater impact on patients' subjective handicap than expected from the motor impairment in a large

Table 1
Comparisons between measured variables.

Comparison	Result
Old HS/New HS*	$R=0.81$; $p<0.0001$
FSS/ODSS*	$R=0.79$; $p<0.0001$
FSS/Old HS*	$R=-0.74$; $p<0.0001$
FSS/New HS*	$R=0.71$; $p<0.0001$
FSS/time since acute GBS	$R=-0.018$; $p=0.93$
FSS/gender	$p=0.83$
FSS/age at acute GBS	$R=0.17$; $p=0.43$
FSS/age at follow-up	$R=0.15$; $p=0.49$
ODSS/Old HS*	$R=0.73$; $p<0.0001$
ODSS /New HS*	$R=0.91$; $p<0.0001$
ODSS/time since acute GBS	$R=0.04$; $p=0.85$
ODSS/gender	$p=0.52$
ODSS/age at acute GBS	$R=0.05$; $p=0.82$
ODSS/age at follow-up	$R=0.04$; $p=0.87$

R = coefficient of correlation.
HS = Hughes score; Old HS = HS at admission for rehabilitation; New HS = HS at follow-up; FSS = Fatigue Severity Scale; ODSS = Overall Disability Sum Score; GBS = Guillain–Barré syndrome.
Statistically significant comparisons are highlighted by *.

proportion of patients, and that this negative effect is maintained for many years following the objective neurological stabilization.

A major part of the difference between the expected dysfunction and the subjective handicap of those patients is explained by complaints of excessive fatigue that are encountered frequently in patients who recovered from GBS.

It is not known yet what is the etiology of fatigue in GBS. Using advanced electrophysiological techniques, a recent study attempted to differentiate between centrally induced fatigue (due to brain dysfunction or disordered mood) and peripherally induced fatigue (due to dysfunction at peripheral nerve, neuromuscular junction and/or muscle level) in well-recovered GBS patients [14]. The results pointed out that an important part of the fatigue recorded in GBS patients might be of central origin. This is a possible explanation for the lack of correlation between the good motor capabilities at neurological examination and the reported poor functioning during daily activities, which depends on additional individual factors, such as mood, arousal and concentration.

Two studies [5,6] evaluated patients with a history of GBS on average 16 months and 5 years previously. 60–80% of them complained of prominent fatigue, independently of muscle strength. In both studies women reported major fatigue more frequently than men. Four similar studies were reported recently, in which patients after GBS were examined on average 6 to 11 years after the acute disease and showed high mean FSS values and a high proportion of patients with FSS > 5, in spite of good neurological recovery [15-18].

In the present study we examined a unique sample of patients who had suffered an attack of GBS more than 20 years previously, the longest follow-up ever reported after GBS. In our study the percentage of patients who reported major disabling fatigue (42%) was slightly lower than in some of the studies mentioned above, possibly due to exclusion of patients with possibly negatively influencing systemic and general neurological conditions. But the percentage of patients with major fatigue is very high also in our series.

We showed also that the subjectively perceived fatigue is well correlated both to the disability at time of the acute disease and at the present examination. These correlations are expected, from a logical point of view, but have not been analyzed systematically previously.

Our data did not reveal any gender-associated differences in the perception of fatigue, as opposed to other series [5,6]. This could be due to the small number of female patients in this study, but possibly also to a better adaptation of women to their functional status with longer follow-up or a lower self-expectation of physical fitness of older women leading to a lower score of self-perceived fatigue.

This is the first study that assessed fatigue in pediatric GBS patients. Generally, the prognosis of recovery in children is considered to be excellent, with a majority of children achieving complete functional recovery within 6 months [19,20]. Our data showed, although in a small number of cases, that many years after the acute GBS attack the mean FSS of the pediatric patients was slightly better than that of adults (3.7 vs 4.5). One 19 years old subject with a severe residual disability (ODSS score 9), however, had extreme fatigue (FSS 6.9).

Recognition of fatigue as a late and disabling symptom after apparent recovery from GBS is important, as physical and behavioral therapy programs may help coping better with the complaints and improve quality of life [21,22].

This study has some inherent limitations: the clinical and electrophysiological data in very old hospitalization charts were at times incomplete or inadequate; the evaluation of the patients at nadir was based on their neurological status at admission to the rehabilitation clinic and not in the acute phase; the present disability and fatigue were based on self report during telephone interviews and not on direct examination; we were able to follow-up only half of the patients, leaving a possible selection bias—the older patients at time of disease might have died before the follow-up study due to non-related diseases; others were lost for follow-up and we cannot assume if those patients were more severely affected and maybe in nursing homes or were the least disabled who returned to full activity with changes in their personal data. We also have not examined the affective status of the subjects, which could have contributed to fatigue. Finally, we do not have data regarding fatigue in a control group, not affected by GBS.

The patients reported in this study did not receive any immunomodulatory treatment during the acute phase, as the effectiveness of plasmapheresis and immunoglobulins had not yet been established. The relevance of the presented data for today's patients is not clear, although in spite of the existent treatments a proportion of patients still remains permanently disabled and shorter follow-up studies published during the last years show similar results in treated patients.

In summary, this is the longest follow-up of patients with GBS ever reported. Almost half (42%) of patients suffered from disabling subjective fatigue more than 20 years after the acute disease. The degree of fatigue correlated to the neurological disability both at admission for in-hospital rehabilitation and at time of the study, although some patients complained of disabling fatigue in spite of making a good neurological recovery.

Abbreviations

FSS	Fatigue severity scale
GBS	Guillain–Barré syndrome
HS	Hughes scale
ODSS	Overall disability sum score

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